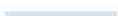


APIs and Interactive Data Apps

Turn notebooks into maintainable projects that obtain and expose data.



Learning outcomes

01

Call HTTP APIs safely and validate responses.

02

Design a JSON endpoint with clear failure states.

03

Build an interactive interface around a data workflow.

The concept system

01

HTTP

02

JSON

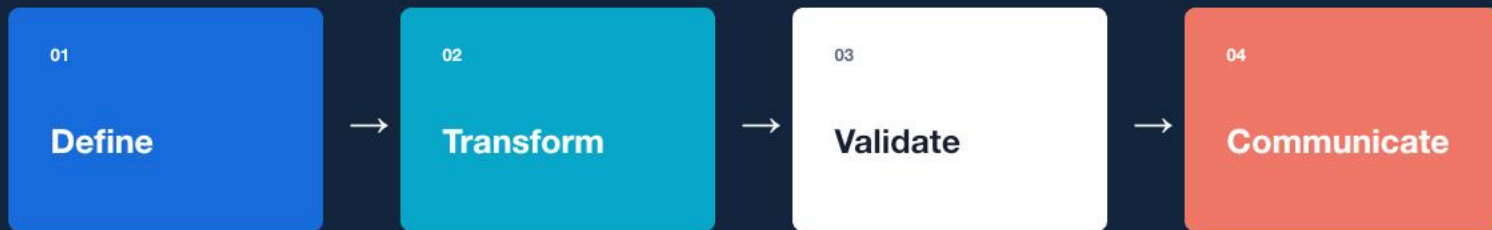
03

API contracts

04

application state

A repeatable reasoning loop



Fetch a public dataset and expose a filtered analytical view.

Guided lab

LAB BRIEF

**Fetch a public dataset and
expose a filtered analytical
view.**

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A documented API client and a small interactive app.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Public Data Explorer

Reliability 30%, architecture 25%, interface 20%, documentation 15%, Git practice 10%.

What should remain after the lesson?

CONCEPT

HTTP + JSON

PRACTICE

Fetch a public dataset and expose a filtered analytical view.

EVIDENCE

A documented API client and a small interactive app.